FOOTING DESIGN CALCULATIONS

Project Name	:	G+2 RB FOOTING DESIGN
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Footing-1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 03:00:45

Definitions:

1.	α	=	Coefficient of Thermal Expansion
2.	ε1	=	Strain at level considered, Calculated ignoring the stiffening of the concrete in tension zone
3.	εm	=	Average steel strain at level considered
4.	ρact	=	Percentage steel provided
5.	ρcrit	=	Percentage steel critical as per code
6.	A	=	Plan area of footing base
7.	Acr	=	Distance from the point considered to the surface of the nearest longitudinal bar
8.	AstPrv	=	Area of tensile reinforcement provided
9.	Astrqd	=	Area of tensile reinforcement required
10.	Asv	=	Area of shear reinforcement required
11.	AsvPrv	=	Area of shear reinforcement provided
12.	A1	=	Bearing area of footing at slope of 1:2
13.	A2	=	C/s Area of column
14.	B	=	Width of footing base
15.	Beff	=	Effective width of footing
16.	BoffB	=	Footing offset along B below column in plan
17.	BoffT	=	Footing offset along B above column in plan
18.	B1	=	Width of sloped footing at top
19.	ColOff	=	Column offset in sloped footing
20.	D	=	Depth of footing
21.	Deff	=	Effective Depth of footing
22.	Df	=	Depth of founding layer
23.	Dw	=	Ground water level
24.	Ec	=	Modulus of elasticity of concrete
25.	Es	=	Modulus of elasticity of steel
26.	Foss	=	Safety factor against sliding
27.	Fosu	=	Safety factor against uplift
28.	Fst	=	Stress in steel
29.	L	=	Length of footing base
30.	Leff	=	Effective length of footing
31.	LoffL	=	Footing offset along L on left side of column

32.	LoffR	=	Footing offset along L on right side of column
33.	L1	=	Length of sloped footing at top
34.	Mx	=	Bending Moment along column D
35.	Mx'	=	Bending Moment along column D due to odd shaped column
36.	Mex	=	Eccentricity moment along column D
37.	Mxx	=	Total Bending Moment along column D (Mx + Mx' + Mex)
38.	My	=	Bending Moment along column B
39.	My'	=	Bending Moment along column B due to odd shaped column
40.	Mey	=	Eccentricity moment along column B
41.	Myy	=	Total Bending Moment along column B (My + My' + Mey)
42.	Muy	=	Factored moment along column B
43.	Muy'	=	Factored Moment along column B due to odd shaped column
44.	Mux	=	Factored moment along column D
45.	Mux'	=	Factored Moment along column D due to odd shaped column
46.	Netdown	=	Net downward load
47.	OB	=	Maximum permissible offset on bottom side of column in plan
48.	OL	=	Maximum permissible offset on left side of column in plan
49.	OR	=	Maximum permissible offset on right side of column in plan
50.	OT	=	Maximum permissible offset on top side of column in plan
51.	P	=	Axial load for footing sizing
52.	P1	=	Soil pressure at corner 1
53.	P2	=	Soil pressure at corner 2
54.	P3	=	Soil pressure at corner 3
55.	P4	=	Soil pressure at corner 4
56.	Pdelta	=	Column Load to be transferred by reinforcement
57.	Pt	=	Calculated percentage tensile reinforcement
58.	Pu	=	Factored axial load
59.	q	=	Surcharge Load
60.	sp	=	Spacing Between bars at outer most layer
61.	SPu	=	Upward Soil Pressure
62.	SPuL	=	Upward Soil Pressure from left
63.	SPuR	=	Upward Soil Pressure from right
64.	SPuB	=	Upward Soil Pressure from bottom
65.	SPuT	=	Upward Soil Pressure from top
66.	Tc	=	Design shear strength of concrete
67.	Temp1	=	Peak Hydration temperature
68.	Temp2	=	Seasonal Temperature Variations
69.	Tv	=	Nominal shear stress

70.	Vu	=	Design shear force
71.	Vus	=	Strength of shear reinforcement
72.	Vx	=	Shear along major axis
73.	Vy	=	Shear along minor axis
74.	Waterpr	=	Upward water pressure
75.	Waterup	=	Upward force due to water
76.	Wcr	=	Surface Crack Width
77.	WFoot	=	Add. Wt. due to difference in concrete and soil density
78.	WFootc	=	Weight of footing and column with concrete density
79.	WSoil	=	Weight of soil covering footing area
80.	Zx	=	Section modulus of footing base along L
81.	Zy	=	Section modulus of footing base along B

Code References:

IS 456

	Parameter		Reference
1.	Ptmax	:	Cl. 26.5.2.1
2.	Ptmin	:	Cl. 26.5.2.1
3.	Pt	:	Cl. 38
4.	Tc	:	Cl. 40.2.1
5.	Tcmax	:	Cl. 40.2.3
6.	Asv	:	Cl. 40.4
7.	Min Shear Reinf	:	Cl. 26.5.1.6
8.	Max Stirrup Spacing	:	Cl. 26.5.1.5
9.	Punching Shear design	:	Cl. 31.6.3
10.	Load transfer	:	Cl. 34.4
11.	Ptnominal	:	Cl. 34.5.2
12.	Crack width calculation	:	Annex-F
13.	Cracking	:	Cl. 35.3.2

BS 8007 (For Early / Initial Thermal Cracking)

1.	Surface Zone for suspended slab	:	Figure A.1
2.	Surface Zone for ground slab	:	Figure A.2
3.	Factors for the calculation of minimum reinforcement	:	Table A.1

Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Footing No	:	FC1	
Column No	:	C1	(230 mm X 450 mm)
Analysis No	:	9	

Concrete Grade	:	M25	N/sqmm
Steel Grade	:	Fe550	N/sqmm
Clear Cover	:	50	mm
Df	:	1.5 M	
Dw	:	4 M	

Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Permissible area of loss of contact	=	0	%

Design cross section by : Average pressure

Footing Type : Pad

Footing Size (L x B x D) : 1450mm X 1250mm X 350mm

Without considering Effect of Water Table

Footing Self Wt., F1	=	15.86	kN
Soil Weight, S1	=	0.00	kN
Total Weight, W1	=	F1 + S1	
	=	15.86	kN
Offset Along L (Loff)	=	500	mm
Offset Along B (Boff)	=	510	mm

Check For Maximum Soil Pressure:

Critical Load Combination	=	[3] : (LOAD 3: DL) +(LOAD 4: LL) -(LOAD 1: EQ X)	
Pcomb	=	256.16	kN
P	=	Pcomb + W1	
P	=	272.02	kN
Mx	=	-2.72	kNm
My	=	-24.08	kNm
P/A	=	150.08	kN/sqm
Mx/Zx	=	-6.22	kN/sqm
My/Zy	=	-63.78	kN/sqm
Maximum Soil Pressure	=	220.08	kN/sqm
Allowable Soil Pressure	=	1.25x200	kN/sqm
	=	250	kN/sqm

Check For Minimum Soil Pressure:

Critical Load Combination	=	[2] : (LOAD 3: DL) + (LOAD 4: LL) +(LOAD 1: EQ X)	
Pcomb	=	116.43	kN
P	=	Pcomb + W1	
P	=	132.29	kN
Mx	=	2.71	kNm
My	=	24.19	kNm
P/A	=	72.99	kN/sqm
Mx/Zx	=	6.2	kN/sqm
My/Zy	=	64.06	kN/sqm
Minimum Soil Pressure	=	2.73	kN/sqm
	>	0	

Design For Bending:

Bottom Reinforcement Along L:

Critical Analysis Load Combination	=	15	
Critical Load Combination	=	[16] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	
Pu	=	305.67	kN
Mux	=	-47.04	kNm
Muy	=	-3.52	kNm
P/A	=	168.65	kN/sqm
Mx/Zx	=	-107.38	kN/sqm
My/Zy	=	-9.32	kN/sqm
Deff	=	292	mm
Beff	=	1250	mm
SPu	=	276.03	kN/sqm
Mu	=	SPu X B X Loff X Loff / 2	
	=	43.13	kNm
Pt	=	0.144	%
Ast Rqd (BM)	=	525	sqmm
Ast Prv (Distributed Across Total Width)	=	6 - T12 @ 300	
	=	679	sqmm

Bottom Reinforcement Along B:

Critical Analysis Load Combination	=	14	
Critical Load Combination	=	[15] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Pu	=	340.77	kN
Mux	=	-5.37	kNm
Muy	=	-36.54	kNm
P/A	=	188.01	kN/sqm
Mx/Zx	=	-12.26	kN/sqm

My/Zy	=	-96.78	kN/sqm
Deff	=	276	mm
Leff	=	1450	mm
SPu	=	284.79	kN/sqm
Mu	=	SPu X L X Boff X Boff / 2	
	=	53.7	kNm
Pt	=	0.152	%
Ast Rqd (BM)	=	609	sqmm
Ast Prv (Distributed Across Total Length)	=	6 - T12 @ 290	
	=	679	sqmm

Design For Shear:

One Way Shear Along L:

Critical Section @ d from Column Face

	=	292	mm
Critical Analysis Load Combination	=	15	
Critical Load Combination	=	[16] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	
Pu	=	305.67	kN
Mux	=	-47.04	kNm
Muy	=	-3.52	kNm
P/A	=	168.65	kN/sqm
Mx/Zx	=	-107.38	kN/sqm
My/Zy	=	-9.32	kN/sqm
Deff	=	292	mm
Beff	=	1250	mm
SPu	=	276.03	kN/sqm
Vu	=	SPu X (Loff - d) X B	
	=	71.77	kN
Tv	=	Vu / (Beff X Deff)	
	=	0.20	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

One Way Shear Along B:

Critical Section @ d from Column Face

	=	276	mm
Critical Analysis Load Combination	=	14	
Critical Load Combination	=	[15] : 1.5 (LOAD 3: DL) - 1.5 (LOAD 1: EQ X)	

Pu	=	340.77	kN
Mux	=	-5.37	kNm
Muy	=	-36.54	kNm
P/A	=	188.01	kN/sqm
Mx/Zx	=	-12.26	kN/sqm
My/Zy	=	-96.78	kN/sqm
Deff	=	276	mm
Leff	=	1450	mm
SPu	=	284.79	kN/sqm
Vu	=	SPu X (Boff - d) X L	
	=	96.63	kN
Tv	=	Vu / (Leff X Deff)	
	=	0.24	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

Design For Punching Shear:

Critical Section @ d/2 from Column Face

	=	142	mm
Critical Analysis Load Combination	=	14	
Critical Load Combination	=	[15] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Pu	=	340.77	kN
Mux	=	-5.37	kNm
Muy	=	-36.54	kNm
P/A	=	188.01	kN/sqm
Mx/Zx	=	-12.26	kN/sqm
My/Zy	=	-96.78	kN/sqm
Deff	=	284	mm
Leff	=	734	mm
Beff	=	514	mm
Punching Perimeter	=	2496	mm
Punching Area	=	708864	sqmm
SPu	=	Average Pressure	
	=	188.01	kN/sqm
Vu	=	SPu X ((L X B)-(Leff X Beff))	
	=	269.84	kN
Tv	=	Vu / Punching Area	
	=	0.381	N/sqmm
Tc	=	1.25	N/sqmm
Tv	<	Tc	

Load Transfer Check For Load Transfer From Column To Footing

Critical Analysis Load Combination	=	14	
Critical Load Combination	=	[15] : 1.5 (LOAD 3: DL) - 1.5 (LOAD 1: EQ X)	
Pu	=	340.77	kN
A2	=	0.1	sqm
A1	=	3.02	sqm
Base Area	=	1.81	sqm
A1	>	Base Area	
Thus, A1	=	1.81	
Modification Factor	=	SquareRoot(A1/A2) < = 2	
SquareRoot(A1/A2)	=	4.2544	
Thus, Modification Factor	=	2	
Concrete Bearing Capacity	=	0.45 X Fck X Modification Factor X Column Area	
	=	2328.75	kN
Concrete Bearing Capacity	>	Pu, Hence Safe.	

Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Footing No	:	FC2	
Column No	:	C2	(230 mm X 450 mm)
Analysis No	:	10	
Concrete Grade	:	M25	N/sqmm
Steel Grade	:	Fe550	N/sqmm
Clear Cover	:	50	mm
Df	:	1.5 M	
Dw	:	4 M	
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Permissible area of loss of contact	=	0	%
Design cross section by	:	Average pressure	
Footing Type	:	Pad	
Footing Size (L x B x D)	:	1500mm X 1300mm X 350mm	

Without considering Effect of Water

Table

Footing Self Wt., F1	=	17.06	kN
Soil Weight, S1	=	0.00	kN
Total Weight, W1	=	F1 + S1	
	=	17.06	kN
Offset Along L (Loff)	=	525	mm
Offset Along B (Boff)	=	535	mm

Check For Maximum Soil Pressure:

Critical Load Combination	=	[5] : (LOAD 3: DL) +(LOAD 4: LL) -(LOAD 2: EQ Z)	
Pcomb	=	304.50	kN
P	=	Pcomb + W1	
P	=	321.57	kN
Mx	=	-33.8	kNm
My	=	1.21	kNm
P/A	=	164.91	kN/sqm
Mx/Zx	=	-69.34	kN/sqm
My/Zy	=	2.87	kN/sqm
Maximum Soil Pressure	=	237.12	kN/sqm
Allowable Soil Pressure	=	1.25x200	kN/sqm
	=	250	kN/sqm

Check For Minimum Soil Pressure:

Critical Load Combination	=	[4] : (LOAD 3: DL) + (LOAD 4: LL) +(LOAD 2: EQ Z)	
Pcomb	=	212.83	kN
P	=	Pcomb + W1	
P	=	229.89	kN
Mx	=	36.34	kNm
My	=	6.27	kNm
P/A	=	117.89	kN/sqm
Mx/Zx	=	74.54	kN/sqm
My/Zy	=	14.85	kN/sqm
Minimum Soil Pressure	=	28.51	kN/sqm
	>	0	

Design For Bending:

Bottom Reinforcement Along L:

Critical Analysis Load Combination	=	15	
Critical Load Combination	=	[16] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	
Pu	=	356.16	kN

Mux	=	-50.45	kNm
Muy	=	0.21	kNm
P/A	=	182.65	kN/sqm
Mx/Zx	=	-103.48	kN/sqm
My/Zy	=	0.49	kN/sqm
Deff	=	292	mm
Beff	=	1300	mm
SPu	=	286.13	kN/sqm
Mu	=	SPu X B X Loff X Loff / 2	
	=	51.26	kNm
Pt	=	0.144	%
Ast Rqd (BM)	=	546	sqmm
Ast Prv (Distributed Across Total Width)	=	6 - T12 @ 300	
	=	679	sqmm

Bottom Reinforcement Along B:

Critical Analysis Load Combination	=	12	
Critical Load Combination	=	[13] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Pu	=	281.85	kN
Mux	=	2.27	kNm
Muy	=	48.38	kNm
P/A	=	144.54	kN/sqm
Mx/Zx	=	4.67	kN/sqm
My/Zy	=	114.52	kN/sqm
Deff	=	276	mm
Leff	=	1500	mm
SPu	=	259.05	kN/sqm
Mu	=	SPu X L X Boff X Boff / 2	
	=	55.61	kNm
Pt	=	0.152	%
Ast Rqd (BM)	=	630	sqmm
Ast Prv (Distributed Across Total Length)	=	6 - T12 @ 300	
	=	679	sqmm

Design For Shear:

One Way Shear Along L:

Critical Section @ d from Column Face			
	=	292	mm
Critical Analysis Load Combination	=	15	

Critical Load Combination	=	[16] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	
Pu	=	356.16	kN
Mux	=	-50.45	kNm
Muy	=	0.21	kNm
P/A	=	182.65	kN/sqm
Mx/Zx	=	-103.48	kN/sqm
My/Zy	=	0.49	kN/sqm
Deff	=	292	mm
Beff	=	1300	mm
SPu	=	286.13	kN/sqm
Vu	=	SPu X (Loff - d) X B	
	=	86.67	kN
Tv	=	Vu / (Beff X Deff)	
	=	0.23	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

One Way Shear Along B:

Critical Section @ d from Column Face

	=	276	mm
Critical Analysis Load Combination	=	12	
Critical Load Combination	=	[13] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Pu	=	281.85	kN
Mux	=	2.27	kNm
Muy	=	48.38	kNm
P/A	=	144.54	kN/sqm
Mx/Zx	=	4.67	kN/sqm
My/Zy	=	114.52	kN/sqm
Deff	=	276	mm
Leff	=	1500	mm
SPu	=	259.05	kN/sqm
Vu	=	SPu X (Boff - d) X L	
	=	100.64	kN
Tv	=	Vu / (Leff X Deff)	
	=	0.24	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

Design For Punching Shear:

Critical Section @ d/2 from Column Face			
	=	142	mm
Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	388	kN
Mux	=	1.9	kNm
Muy	=	5.61	kNm
P/A	=	198.97	kN/sqm
Mx/Zx	=	3.9	kN/sqm
My/Zy	=	13.29	kN/sqm
Deff	=	284	mm
Leff	=	734	mm
Beff	=	514	mm
Punching Perimeter	=	2496	mm
Punching Area	=	708864	sqmm
SPu	=	Average Pressure	
	=	198.97	kN/sqm
Vu	=	SPu X ((L X B)-(Leff X Beff))	
	=	312.93	kN
Tv	=	Vu / Punching Area	
	=	0.441	N/sqmm
Tc	=	1.25	N/sqmm
Tv	<	Tc	

Load Transfer Check For Load Transfer From Column To Footing

Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	388	kN
A2	=	0.1	sqm
A1	=	3.02	sqm
Base Area	=	1.95	sqm
A1	>	Base Area	
Thus, A1	=	1.95	
Modification Factor	=	SquareRoot(A1/A2) < = 2	
SquareRoot(A1/A2)	=	4.4159	
Thus, Modification Factor	=	2	
Concrete Bearing Capacity	=	0.45 X Fck X Modification Factor X Column Area	
	=	2328.75	kN
Concrete Bearing Capacity	>	Pu, Hence Safe.	

Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Footing No	:	FC3	
Column No	:	C3	(230 mm X 450 mm)
Analysis No	:	11	
Concrete Grade	:	M25	N/sqmm
Steel Grade	:	Fe550	N/sqmm
Clear Cover	:	50	mm
Df	:	1.5 M	
Dw	:	4 M	
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Permissible area of loss of contact	=	0	%
Design cross section by	:	Average pressure	

Footing Type	:	Pad	
Footing Size (L x B x D)	:	1700mm X 1500mm X 425mm	
Without considering Effect of Water Table			
Footing Self Wt., F1	=	27.09	kN
Soil Weight, S1	=	0.00	kN
Total Weight, W1	=	F1 + S1	
	=	27.09	kN
Offset Along L (Loff)	=	625	mm
Offset Along B (Boff)	=	635	mm

Check For Maximum Soil Pressure:

Critical Load Combination	=	[5] : (LOAD 3: DL) +(LOAD 4: LL) -(LOAD 2: EQ Z)	
Pcomb	=	432.46	kN
P	=	Pcomb + W1	
P	=	459.55	kN
Mx	=	-43.2	kNm
My	=	-1.26	kNm
P/A	=	180.22	kN/sqm
Mx/Zx	=	-59.79	kN/sqm

My/Zy	=	-1.97	kN/sqm
Maximum Soil Pressure	=	241.98	kN/sqm
Allowable Soil Pressure	=	1.25x200	kN/sqm
	=	250	kN/sqm

Check For Minimum Soil Pressure:

Critical Load Combination	=	[4] : (LOAD 3: DL) + (LOAD 4: LL) +(LOAD 2: EQ Z)	
Pcomb	=	310.36	kN
P	=	Pcomb + W1	
P	=	337.45	kN
Mx	=	41.76	kNm
My	=	3.64	kNm
P/A	=	132.33	kN/sqm
Mx/Zx	=	57.8	kN/sqm
My/Zy	=	5.7	kN/sqm
Minimum Soil Pressure	=	68.83	kN/sqm
	>	0	

Design For Bending:

Bottom Reinforcement Along L:

Critical Analysis Load Combination	=	15	
Critical Load Combination	=	[16] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	
Pu	=	539.58	kN
Mux	=	-65.09	kNm
Muy	=	-2.32	kNm
P/A	=	211.6	kN/sqm
Mx/Zx	=	-90.09	kN/sqm
My/Zy	=	-3.64	kN/sqm
Deff	=	367	mm
Beff	=	1500	mm
SPu	=	301.69	kN/sqm
Mu	=	SPu X B X Loff X Loff / 2	
	=	88.39	kNm
Pt	=	0.139	%
Ast Rqd (BM)	=	765	sqmm
Ast Prv (Distributed Across Total Width)	=	7 - T12 @ 250	
	=	792	sqmm

Bottom Reinforcement Along B:

Critical Analysis Load Combination	=	12	

Critical Load Combination	=	[13] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Pu	=	558.79	kN
Mux	=	-5.2	kNm
Muy	=	37.46	kNm
P/A	=	219.13	kN/sqm
Mx/Zx	=	-7.19	kN/sqm
My/Zy	=	58.77	kN/sqm
Deff	=	351	mm
Leff	=	1700	mm
SPu	=	277.9	kN/sqm
Mu	=	SPu X L X Boff X Boff / 2	
	=	95.25	kNm
Pt	=	0.145	%
Ast Rqd (BM)	=	867	sqmm
Ast Prv (Distributed Across Total Length)	=	8 - T12 @ 245	
	=	905	sqmm

Design For Shear:

One Way Shear Along L:

Critical Section @ d from Column Face			
	=	367	mm
Critical Analysis Load Combination	=	15	
Critical Load Combination	=	[16] : 1.5 (LOAD 3: DL) -1.5 (LOAD 2: EQ Z)	
Pu	=	539.58	kN
Mux	=	-65.09	kNm
Muy	=	-2.32	kNm
P/A	=	211.6	kN/sqm
Mx/Zx	=	-90.09	kN/sqm
My/Zy	=	-3.64	kN/sqm
Deff	=	367	mm
Beff	=	1500	mm
SPu	=	301.69	kN/sqm
Vu	=	SPu X (Loff - d) X B	
	=	116.75	kN
Tv	=	Vu / (Beff X Deff)	
	=	0.21	N/sqmm
Tc	=	0.28	N/sqmm
Tv	<	Tc	

One Way Shear Along B:

Critical Section @ d from Column Face

	=	351	mm
Critical Analysis Load Combination	=	12	
Critical Load Combination	=	[13] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Pu	=	558.79	kN
Mux	=	-5.2	kNm
Muy	=	37.46	kNm
P/A	=	219.13	kN/sqm
Mx/Zx	=	-7.19	kN/sqm
My/Zy	=	58.77	kN/sqm
Deff	=	351	mm
Leff	=	1700	mm
SPu	=	277.9	kN/sqm
Vu	=	SPu X (Boff - d) X L	
	=	134.17	kN
Tv	=	Vu / (Leff X Deff)	
	=	0.22	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

Design For Punching Shear:

Critical Section @ d/2 from Column Face

	=	180	mm
Critical Analysis Load Combination	=	12	
Critical Load Combination	=	[13] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Pu	=	558.79	kN
Mux	=	-5.2	kNm
Muy	=	37.46	kNm
P/A	=	219.13	kN/sqm
Mx/Zx	=	-7.19	kN/sqm
My/Zy	=	58.77	kN/sqm
Deff	=	359	mm
Leff	=	809	mm
Beff	=	589	mm
Punching Perimeter	=	2796	mm
Punching Area	=	1003764	sqmm
SPu	=	Average Pressure	

	=	219.13	kN/sqm
Vu	=	SPu X ((L X B)-(Leff X Beff))	
	=	454.37	kN
Tv	=	Vu / Punching Area	
	=	0.453	N/sqmm
Tc	=	1.25	N/sqmm
Tv	<	Tc	

Load Transfer Check For Load Transfer From Column To Footing

Critical Analysis Load Combination	=	12	
Critical Load Combination	=	[13] : 1.5 (LOAD 3: DL) +1.5 (LOAD 1: EQ X)	
Pu	=	558.79	kN
A2	=	0.1	sqm
A1	=	4.15	sqm
Base Area	=	2.55	sqm
A1	>	Base Area	
Thus, A1	=	2.55	
Modification Factor	=	SquareRoot(A1/A2) < = 2	
SquareRoot(A1/A2)	=	5.0498	
Thus, Modification Factor	=	2	
Concrete Bearing Capacity	=	0.45 X Fck X Modification Factor X Column Area	
	=	2328.75	kN
Concrete Bearing Capacity	>	Pu, Hence Safe.	

Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Footing No	:	FC4	
Column No	:	C4	(230 mm X 450 mm)
Analysis No	:	29	
Concrete Grade	:	M25	N/sqmm
Steel Grade	:	Fe550	N/sqmm
Clear Cover	:	50	mm
Df	:	1.5 M	
Dw	:	4 M	
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Permissible area of loss of contact	=	0	%

Design cross section by	:	Average pressure	
Footing Type	:	Pad	
Footing Size (L x B x D)	:	1600mm X 1400mm X 375mm	
Without considering Effect of Water Table			
Footing Self Wt., F1	=	21.00	kN
Soil Weight, S1	=	0.00	kN
Total Weight, W1	=	F1 + S1	
	=	21	kN
Offset Along L (Loff)	=	575	mm
Offset Along B (Boff)	=	585	mm

Check For Maximum Soil Pressure:

Critical Load Combination	=	[3] : (LOAD 3: DL) +(LOAD 4: LL) -(LOAD 1: EQ X)	
Pcomb	=	377.14	kN
P	=	Pcomb + W1	
P	=	398.14	kN
Mx	=	3.73	kNm
My	=	-25.68	kNm
P/A	=	177.74	kN/sqm
Mx/Zx	=	6.25	kN/sqm
My/Zy	=	-49.12	kN/sqm
Maximum Soil Pressure	=	233.11	kN/sqm
Allowable Soil Pressure	=	1.25x200	kN/sqm
	=	250	kN/sqm

Check For Minimum Soil Pressure:

Critical Load Combination	=	[2] : (LOAD 3: DL) + (LOAD 4: LL) +(LOAD 1: EQ X)	
Pcomb	=	233.16	kN
P	=	Pcomb + W1	
P	=	254.16	kN
Mx	=	9.95	kNm
My	=	27.34	kNm
P/A	=	113.46	kN/sqm
Mx/Zx	=	16.65	kN/sqm
My/Zy	=	52.31	kN/sqm
Minimum Soil Pressure	=	44.5	kN/sqm

	>	0	

Design For Bending:

Bottom Reinforcement Along L:

Critical Analysis Load Combination	=	13	
Critical Load Combination	=	[14] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	
Pu	=	356.48	kN
Mux	=	59.3	kNm
Muy	=	2.34	kNm
P/A	=	159.14	kN/sqm
Mx/Zx	=	99.28	kN/sqm
My/Zy	=	4.48	kN/sqm
Deff	=	317	mm
Beff	=	1400	mm
SPu	=	258.43	kN/sqm
Mu	=	SPu X B X Loff X Loff / 2	
	=	59.81	kNm
Pt	=	0.142	%
Ast Rqd (BM)	=	630	sqmm
Ast Prv (Distributed Across Total Width)	=	6 - T12 @ 280	
	=	679	sqmm

Bottom Reinforcement Along B:

Critical Analysis Load Combination	=	14	
Critical Load Combination	=	[15] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Pu	=	459.31	kN
Mux	=	2.05	kNm
Muy	=	-38.34	kNm
P/A	=	205.05	kN/sqm
Mx/Zx	=	3.43	kN/sqm
My/Zy	=	-73.35	kN/sqm
Deff	=	301	mm
Leff	=	1600	mm
SPu	=	278.4	kN/sqm
Mu	=	SPu X L X Boff X Boff / 2	
	=	76.22	kNm
Pt	=	0.15	%
Ast Rqd (BM)	=	720	sqmm
Ast Prv (Distributed Across Total Length)	=	7 - T12 @ 270	
	=	792	sqmm

Design For Shear:

One Way Shear Along L:

Critical Section @ d from Column Face

	=	317	mm
Critical Analysis Load Combination	=	13	
Critical Load Combination	=	[14] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	
Pu	=	356.48	kN
Mux	=	59.3	kNm
Muy	=	2.34	kNm
P/A	=	159.14	kN/sqm
Mx/Zx	=	99.28	kN/sqm
My/Zy	=	4.48	kN/sqm
Deff	=	317	mm
Beff	=	1400	mm
SPu	=	258.43	kN/sqm
Vu	=	SPu X (Loff - d) X B	
	=	93.34	kN
Tv	=	Vu / (Beff X Deff)	
	=	0.21	N/sqmm
Tc	=	0.28	N/sqmm
Tv	<	Tc	

One Way Shear Along B:

Critical Section @ d from Column Face

	=	301	mm
Critical Analysis Load Combination	=	14	
Critical Load Combination	=	[15] : 1.5 (LOAD 3: DL) - 1.5 (LOAD 1: EQ X)	
Pu	=	459.31	kN
Mux	=	2.05	kNm
Muy	=	-38.34	kNm
P/A	=	205.05	kN/sqm
Mx/Zx	=	3.43	kN/sqm
My/Zy	=	-73.35	kN/sqm
Deff	=	301	mm
Leff	=	1600	mm
SPu	=	278.4	kN/sqm
Vu	=	SPu X (Boff - d) X L	

	=	126.50	kN
Tv	=	Vu / (Leff X Deff)	
	=	0.26	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

Design For Punching Shear:

Critical Section @ d/2 from Column Face

	=	154	mm

Critical Analysis Load Combination	=	14	
Critical Load Combination	=	[15] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Pu	=	459.31	kN
Mux	=	2.05	kNm
Muy	=	-38.34	kNm
P/A	=	205.05	kN/sqm
Mx/Zx	=	3.43	kN/sqm
My/Zy	=	-73.35	kN/sqm
Deff	=	309	mm
Leff	=	759	mm
Beff	=	539	mm
Punching Perimeter	=	2596	mm
Punching Area	=	802164	sqmm
SPu	=	Average Pressure	
	=	205.05	kN/sqm
Vu	=	SPu X ((L X B)-(Leff X Beff))	
	=	375.42	kN
Tv	=	Vu / Punching Area	
	=	0.468	N/sqmm
Tc	=	1.25	N/sqmm
Tv	<	Tc	

Load Transfer Check For Load Transfer From Column To Footing

Critical Analysis Load Combination	=	14	
Critical Load Combination	=	[15] : 1.5 (LOAD 3: DL) - 1.5 (LOAD 1: EQ X)	
Pu	=	459.31	kN
A2	=	0.1	sqm
A1	=	3.37	sqm
Base Area	=	2.24	sqm
A1	>	Base Area	
Thus, A1	=	2.24	

Modification Factor	=	SquareRoot(A1/A2) < = 2	
SquareRoot(A1/A2)	=	4.7329	
Thus, Modification Factor	=	2	
Concrete Bearing Capacity	=	0.45 X Fck X Modification Factor X Column Area	
	=	2328.75	kN
Concrete Bearing Capacity	>	Pu, Hence Safe.	

Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Footing No	:	FC5	
Column No	:	C5	(230 mm X 450 mm)
Analysis No	:	30	
Concrete Grade	:	M25	N/sqmm
Steel Grade	:	Fe550	N/sqmm
Clear Cover	:	50	mm
Df	:	1.5 M	
Dw	:	4 M	
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Permissible area of loss of contact	=	0	%
Design cross section by	:	Average pressure	

Footing Type	:	Pad	
Footing Size (L x B x D)	:	1600mm X 1400mm X 375mm	
Without considering Effect of Water Table			
Footing Self Wt., F1	=	21.00	kN
Soil Weight, S1	=	0.00	kN
Total Weight, W1	=	F1 + S1	
	=	21	kN
Offset Along L (Loff)	=	575	mm
Offset Along B (Boff)	=	585	mm

Check For Maximum Soil Pressure:

Critical Load Combination	=	[4] : (LOAD 3: DL) +(LOAD 4: LL) +(LOAD 2: EQ Z)

Pcomb	=	299.63	kN
P	=	Pcomb + W1	
P	=	320.63	kN
Mx	=	47.54	kNm
My	=	5.13	kNm
P/A	=	143.14	kN/sqm
Mx/Zx	=	79.58	kN/sqm
My/Zy	=	9.81	kN/sqm
Maximum Soil Pressure	=	232.54	kN/sqm
Allowable Soil Pressure	=	1.25x200	kN/sqm
	=	250	kN/sqm

Check For Minimum Soil Pressure:

Critical Load Combination	=	[4] : (LOAD 3: DL) + (LOAD 4: LL) +(LOAD 2: EQ Z)	
Pcomb	=	299.63	kN
P	=	Pcomb + W1	
P	=	320.63	kN
Mx	=	47.54	kNm
My	=	5.13	kNm
P/A	=	143.14	kN/sqm
Mx/Zx	=	79.58	kN/sqm
My/Zy	=	9.81	kN/sqm
Minimum Soil Pressure	=	53.74	kN/sqm
	>	0	

Design For Bending:

Bottom Reinforcement Along L:

Critical Analysis Load Combination	=	8	
Critical Load Combination	=	[9] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)	
Pu	=	359.56	kN
Mux	=	57.04	kNm
Muy	=	6.16	kNm
P/A	=	160.52	kN/sqm
Mx/Zx	=	95.5	kN/sqm
My/Zy	=	11.78	kN/sqm
Deff	=	317	mm
Beff	=	1400	mm
SPu	=	256.02	kN/sqm
Mu	=	SPu X B X Loff X Loff / 2	

	=	59.25	kNm
Pt	=	0.142	%
Ast Rqd (BM)	=	630	sqmm
Ast Prv (Distributed Across Total Width)	=	6 - T12 @ 280	
	=	679	sqmm

Bottom Reinforcement Along B:

Critical Analysis Load Combination	=	7	
Critical Load Combination	=	[8] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)	
Pu	=	353.38	kN
Mux	=	8.57	kNm
Muy	=	44.13	kNm
P/A	=	157.76	kN/sqm
Mx/Zx	=	14.35	kN/sqm
My/Zy	=	84.42	kN/sqm
Deff	=	301	mm
Leff	=	1600	mm
SPu	=	242.18	kN/sqm
Mu	=	SPu X L X Boff X Boff / 2	
	=	66.3	kNm
Pt	=	0.15	%
Ast Rqd (BM)	=	720	sqmm
Ast Prv (Distributed Across Total Length)	=	7 - T12 @ 270	
	=	792	sqmm

Design For Shear:

One Way Shear Along L:

Critical Section @ d from Column Face			
	=	317	mm
Critical Analysis Load Combination	=	8	
Critical Load Combination	=	[9] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)	
Pu	=	359.56	kN
Mux	=	57.04	kNm
Muy	=	6.16	kNm
P/A	=	160.52	kN/sqm
Mx/Zx	=	95.5	kN/sqm
My/Zy	=	11.78	kN/sqm
Deff	=	317	mm

Beff	=	1400	mm
SPu	=	256.02	kN/sqm
Vu	=	SPu X (Loff - d) X B	
	=	92.47	kN
Tv	=	Vu / (Beff X Deff)	
	=	0.21	N/sqmm
Tc	=	0.28	N/sqmm
Tv	<	Tc	

One Way Shear Along B:

Critical Section @ d from Column Face

	=	301	mm
Critical Analysis Load Combination	=	7	
Critical Load Combination	=	[8] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)	
Pu	=	353.38	kN
Mux	=	8.57	kNm
Muy	=	44.13	kNm
P/A	=	157.76	kN/sqm
Mx/Zx	=	14.35	kN/sqm
My/Zy	=	84.42	kN/sqm

Deff	=	301	mm
Leff	=	1600	mm
SPu	=	242.18	kN/sqm
Vu	=	SPu X (Boff - d) X L	
	=	110.05	kN
Tv	=	Vu / (Leff X Deff)	
	=	0.23	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

Design For Punching Shear:

Critical Section @ d/2 from Column Face

	=	154	mm

Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	445.97	kN
Mux	=	10.58	kNm
Muy	=	6.56	kNm
P/A	=	199.09	kN/sqm

Mx/Zx	=	17.71	kN/sqm
My/Zy	=	12.56	kN/sqm
Deff	=	309	mm
Leff	=	759	mm
Beff	=	539	mm
Punching Perimeter	=	2596	mm
Punching Area	=	802164	sqmm
SPu	=	Average Pressure	
	=	199.09	kN/sqm
Vu	=	SPu X ((L X B)-(Leff X Beff))	
	=	364.52	kN
Tv	=	Vu / Punching Area	
	=	0.454	N/sqmm
Tc	=	1.25	N/sqmm
Tv	<	Tc	

Load Transfer Check For Load Transfer From Column To Footing

Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	445.97	kN
A2	=	0.1	sqm
A1	=	3.37	sqm
Base Area	=	2.24	sqm
A1	>	Base Area	
Thus, A1	=	2.24	
Modification Factor	=	SquareRoot(A1/A2) < = 2	
SquareRoot(A1/A2)	=	4.7329	
Thus, Modification Factor	=	2	
Concrete Bearing Capacity	=	0.45 X Fck X Modification Factor X Column Area	
	=	2328.75	kN
Concrete Bearing Capacity	>	Pu, Hence Safe.	

Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Footing No	:	FC6	
Column No	:	C6	(230 mm X 450 mm)
Analysis No	:	31	
Concrete Grade	:	M25	N/sqmm
Steel Grade	:	Fe550	N/sqmm
Clear Cover	:	50	mm

Df	:	1.5 M	
Dw	:	4 M	
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Permissible area of loss of contact	=	0	%
Design cross section by	:	Average pressure	
Footing Type	:	Pad	
Footing Size (L x B x D)	:	1900mm X 1700mm X 450mm	

Without considering Effect of Water Table

Footing Self Wt., F1	=	36.34	kN
Soil Weight, S1	=	0.00	kN
Total Weight, W1	=	F1 + S1	
	=	36.34	kN
Offset Along L (Loff)	=	725	mm
Offset Along B (Boff)	=	735	mm

Check For Maximum Soil Pressure:

Critical Load Combination	=	[4] : (LOAD 3: DL) +(LOAD 4: LL) +(LOAD 2: EQ Z)	
Pcomb	=	574.96	kN
P	=	Pcomb + W1	
P	=	611.30	kN
Mx	=	56.25	kNm
My	=	-1.79	kNm
P/A	=	189.26	kN/sqm
Mx/Zx	=	55	kN/sqm
My/Zy	=	-1.95	kN/sqm
Maximum Soil Pressure	=	246.21	kN/sqm
Allowable Soil Pressure	=	1.25x200	kN/sqm
	=	250	kN/sqm

Check For Minimum Soil Pressure:

Critical Load Combination	=	[3] : (LOAD 3: DL) + (LOAD 4: LL) -(LOAD 1: EQ X)	
Pcomb	=	487.22	kN

P	=	Pcomb + W1	
P	=	523.56	kN
Mx	=	10.68	kNm
My	=	-28.83	kNm
P/A	=	162.09	kN/sqm
Mx/Zx	=	10.44	kN/sqm
My/Zy	=	-31.5	kN/sqm
Minimum Soil Pressure	=	120.15	kN/sqm
	>	0	

Design For Bending:

Bottom Reinforcement Along L:

Critical Analysis Load Combination	=	8	
Critical Load Combination	=	[9] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)	
Pu	=	689.96	kN
Mux	=	67.51	kNm
Muy	=	-2.14	kNm
P/A	=	213.61	kN/sqm
Mx/Zx	=	66	kN/sqm
My/Zy	=	-2.34	kN/sqm
Deff	=	392	mm
Beff	=	1700	mm
SPu	=	279.61	kN/sqm
Mu	=	SPu X B X Loff X Loff / 2	
	=	124.92	kNm
Pt	=	0.138	%
Ast Rqd (BM)	=	918	sqmm
Ast Prv (Distributed Across Total Width)	=	9 - T12 @ 215	
	=	1018	sqmm

Bottom Reinforcement Along B:

Critical Analysis Load Combination	=	7	
Critical Load Combination	=	[8] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)	
Pu	=	761.58	kN
Mux	=	5.87	kNm
Muy	=	29.08	kNm
P/A	=	235.78	kN/sqm
Mx/Zx	=	5.74	kN/sqm
My/Zy	=	31.78	kN/sqm

Deff	=	376	mm
Leff	=	1900	mm
SPu	=	267.56	kN/sqm
Mu	=	SPu X L X Boff X Boff / 2	
	=	137.32	kNm
Pt	=	0.144	%
Ast Rqd (BM)	=	1026	sqmm
Ast Prv (Distributed Across Total Length)	=	10 - T12 @ 215	
	=	1131	sqmm

Design For Shear:

One Way Shear Along L:

Critical Section @ d from Column Face

	=	392	mm
Critical Analysis Load Combination	=	8	
Critical Load Combination	=	[9] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)	
Pu	=	689.96	kN
Mux	=	67.51	kNm
Muy	=	-2.14	kNm
P/A	=	213.61	kN/sqm
Mx/Zx	=	66	kN/sqm
My/Zy	=	-2.34	kN/sqm

Deff	=	392	mm
Beff	=	1700	mm
SPu	=	279.61	kN/sqm
Vu	=	SPu X (Loff - d) X B	
	=	158.29	kN
Tv	=	Vu / (Beff X Deff)	
	=	0.24	N/sqmm
Tc	=	0.28	N/sqmm
Tv	<	Tc	

One Way Shear Along B:

Critical Section @ d from Column Face

	=	376	mm
Critical Analysis Load Combination	=	7	
Critical Load Combination	=	[8] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)	

Pu	=	761.58	kN
Mux	=	5.87	kNm
Muy	=	29.08	kNm
P/A	=	235.78	kN/sqm
Mx/Zx	=	5.74	kN/sqm
My/Zy	=	31.78	kN/sqm
Deff	=	376	mm
Leff	=	1900	mm
SPu	=	267.56	kN/sqm
Vu	=	SPu X (Boff - d) X L	
	=	182.50	kN
Tv	=	Vu / (Leff X Deff)	
	=	0.26	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

Design For Punching Shear:

Critical Section @ d/2 from Column Face

	=	192	mm
Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	841.4	kN
Mux	=	11.68	kNm
Muy	=	-3.45	kNm
P/A	=	260.5	kN/sqm
Mx/Zx	=	11.42	kN/sqm
My/Zy	=	-3.77	kN/sqm
Deff	=	384	mm
Leff	=	834	mm
Beff	=	614	mm
Punching Perimeter	=	2896	mm
Punching Area	=	1112064	sqmm
SPu	=	Average Pressure	
	=	260.5	kN/sqm
Vu	=	SPu X ((L X B)-(Leff X Beff))	
	=	708.01	kN
Tv	=	Vu / Punching Area	
	=	0.637	N/sqmm
Tc	=	1.25	N/sqmm
Tv	<	Tc	

Load Transfer Check For Load Transfer From Column To Footing

Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	841.4	kN
A2	=	0.1	sqm
A1	=	4.57	sqm
Base Area	=	3.23	sqm
A1	>	Base Area	
Thus, A1	=	3.23	
Modification Factor	=	SquareRoot(A1/A2) < = 2	
SquareRoot(A1/A2)	=	5.6833	
Thus, Modification Factor	=	2	
Concrete Bearing Capacity	=	0.45 X Fck X Modification Factor X Column Area	
	=	2328.75	kN
Concrete Bearing Capacity	>	Pu, Hence Safe.	

Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Footing No	:	FC7	
Column No	:	C7	(230 mm X 450 mm)
Analysis No	:	49	
Concrete Grade	:	M25	N/sqmm
Steel Grade	:	Fe550	N/sqmm
Clear Cover	:	50	mm
Df	:	1.5 M	
Dw	:	4 M	
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Permissible area of loss of contact	=	0	%
Design cross section by	:	Average pressure	
Footing Type	:	Pad	
Footing Size (L x B x D)	:	1550mm X 1350mm X 375mm	

Without considering Effect of Water

Table

Footing Self Wt., F1	=	19.62	kN
Soil Weight, S1	=	0.00	kN
Total Weight, W1	=	F1 + S1	
	=	19.62	kN
Offset Along L (Loff)	=	550	mm
Offset Along B (Boff)	=	560	mm

Check For Maximum Soil Pressure:

Critical Load Combination	=	[3] : (LOAD 3: DL) +(LOAD 4: LL) -(LOAD 1: EQ X)	
Pcomb	=	364.26	kN
P	=	Pcomb + W1	
P	=	383.87	kN
Mx	=	4.58	kNm
My	=	-27.2	kNm
P/A	=	183.45	kN/sqm
Mx/Zx	=	8.47	kN/sqm
My/Zy	=	-57.77	kN/sqm
Maximum Soil Pressure	=	249.68	kN/sqm
Allowable Soil Pressure	=	1.25x200	kN/sqm
	=	250	kN/sqm

Check For Minimum Soil Pressure:

Critical Load Combination	=	[2] : (LOAD 3: DL) + (LOAD 4: LL) +(LOAD 1: EQ X)	
Pcomb	=	207.40	kN
P	=	Pcomb + W1	
P	=	227.02	kN
Mx	=	10.79	kNm
My	=	28.61	kNm
P/A	=	108.49	kN/sqm
Mx/Zx	=	19.97	kN/sqm
My/Zy	=	60.76	kN/sqm
Minimum Soil Pressure	=	27.76	kN/sqm
	>	0	

Design For Bending:

Bottom Reinforcement Along L:

Critical Analysis Load Combination	=	13	
Critical Load Combination	=	[14] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	
Pu	=	296.98	kN

Mux	=	61.1	kNm
Muy	=	0.82	kNm
P/A	=	141.92	kN/sqm
Mx/Zx	=	113.03	kN/sqm
My/Zy	=	1.73	kN/sqm
Deff	=	317	mm
Beff	=	1350	mm
SPu	=	254.96	kN/sqm
Mu	=	SPu X B X Loff X Loff / 2	
	=	52.06	kNm
Pt	=	0.142	%
Ast Rqd (BM)	=	608	sqmm
Ast Prv (Distributed Across Total Width)	=	6 - T12 @ 270	
	=	679	sqmm

Bottom Reinforcement Along B:

Critical Analysis Load Combination	=	14	
Critical Load Combination	=	[15] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Pu	=	434.32	kN
Mux	=	3.35	kNm
Muy	=	-40.33	kNm
P/A	=	207.56	kN/sqm
Mx/Zx	=	6.19	kN/sqm
My/Zy	=	-85.67	kN/sqm
Deff	=	301	mm
Leff	=	1550	mm
SPu	=	293.23	kN/sqm
Mu	=	SPu X L X Boff X Boff / 2	
	=	71.27	kNm
Pt	=	0.15	%
Ast Rqd (BM)	=	698	sqmm
Ast Prv (Distributed Across Total Length)	=	7 - T12 @ 260	
	=	792	sqmm

Design For Shear:

One Way Shear Along L:

Critical Section @ d from Column Face			
	=	317	mm
Critical Analysis Load Combination	=	13	

Critical Load Combination	=	[14] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	
Pu	=	296.98	kN
Mux	=	61.1	kNm
Muy	=	0.82	kNm
P/A	=	141.92	kN/sqm
Mx/Zx	=	113.03	kN/sqm
My/Zy	=	1.73	kN/sqm
Deff	=	317	mm
Beff	=	1350	mm
SPu	=	254.96	kN/sqm
Vu	=	SPu X (Loff - d) X B	
	=	80.20	kN
Tv	=	Vu / (Beff X Deff)	
	=	0.19	N/sqmm
Tc	=	0.28	N/sqmm
Tv	<	Tc	

One Way Shear Along B:

Critical Section @ d from Column Face

	=	301	mm
Critical Analysis Load Combination	=	14	
Critical Load Combination	=	[15] : 1.5 (LOAD 3: DL) - 1.5 (LOAD 1: EQ X)	
Pu	=	434.32	kN
Mux	=	3.35	kNm
Muy	=	-40.33	kNm
P/A	=	207.56	kN/sqm
Mx/Zx	=	6.19	kN/sqm
My/Zy	=	-85.67	kN/sqm
Deff	=	301	mm
Leff	=	1550	mm
SPu	=	293.23	kN/sqm
Vu	=	SPu X (Boff - d) X L	
	=	117.72	kN
Tv	=	Vu / (Leff X Deff)	
	=	0.25	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

Design For Punching Shear:

Critical Section @ d/2 from Column Face			
	=	154	mm
Critical Analysis Load Combination	=	9	
Critical Load Combination	=	[10] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) - 1.2 (LOAD 1: EQ X)	
Pu	=	437.11	kN
Mux	=	5.49	kNm
Muy	=	-32.64	kNm
P/A	=	208.89	kN/sqm
Mx/Zx	=	10.16	kN/sqm
My/Zy	=	-69.32	kN/sqm
Deff	=	309	mm
Leff	=	759	mm
Beff	=	539	mm
Punching Perimeter	=	2596	mm
Punching Area	=	802164	sqmm
SPu	=	Average Pressure	
	=	208.89	kN/sqm
Vu	=	SPu X ((L X B)-(Leff X Beff))	
	=	351.65	kN
Tv	=	Vu / Punching Area	
	=	0.438	N/sqmm
Tc	=	1.25	N/sqmm
Tv	<	Tc	

Load Transfer Check For Load Transfer From Column To Footing

Critical Analysis Load Combination	=	9	
Critical Load Combination	=	[10] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) -1.2 (LOAD 1: EQ X)	
Pu	=	437.11	kN
A2	=	0.1	sqm
A1	=	3.37	sqm
Base Area	=	2.09	sqm
A1	>	Base Area	
Thus, A1	=	2.09	
Modification Factor	=	SquareRoot(A1/A2) < = 2	
SquareRoot(A1/A2)	=	4.5717	
Thus, Modification Factor	=	2	
Concrete Bearing Capacity	=	0.45 X Fck X Modification Factor X Column Area	
	=	2328.75	kN

Concrete Bearing Capacity	>	Pu, Hence Safe.

Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Footing No	:	FC8	
Column No	:	C8	(230 mm X 530 mm)
Analysis No	:	50	
Concrete Grade	:	M25	N/sqmm
Steel Grade	:	Fe550	N/sqmm
Clear Cover	:	50	mm
Df	:	1.5 M	
Dw	:	4 M	
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Permissible area of loss of contact	=	0	%
Design cross section by	:	Average pressure	

Footing Type	:	Pad	
Footing Size (L x B x D)	:	1700mm X 1400mm X 375mm	
Without considering Effect of Water Table			
Footing Self Wt., F1	=	22.31	kN
Soil Weight, S1	=	0.00	kN
Total Weight, W1	=	F1 + S1	
	=	22.31	kN
Offset Along L (Loff)	=	585	mm
Offset Along B (Boff)	=	585	mm

Check For Maximum Soil Pressure:

Critical Load Combination	=	[4] : (LOAD 3: DL) +(LOAD 4: LL) +(LOAD 2: EQ Z)	
Pcomb	=	273.72	kN
P	=	Pcomb + W1	
P	=	296.03	kN
Mx	=	70.18	kNm
My	=	4.44	kNm
P/A	=	124.38	kN/sqm

Mx/Zx	=	104.08	kN/sqm
My/Zy	=	8	kN/sqm
Maximum Soil Pressure	=	236.46	kN/sqm
Allowable Soil Pressure	=	1.25x200	kN/sqm
	=	250	kN/sqm

Check For Minimum Soil Pressure:

Critical Load Combination	=	[4] : (LOAD 3: DL) + (LOAD 4: LL) +(LOAD 2: EQ Z)	
Pcomb	=	273.72	kN
P	=	Pcomb + W1	
P	=	296.03	kN
Mx	=	70.18	kNm
My	=	4.44	kNm
P/A	=	124.38	kN/sqm
Mx/Zx	=	104.08	kN/sqm
My/Zy	=	8	kN/sqm
Minimum Soil Pressure	=	12.3	kN/sqm
	>	0	

Design For Bending:

Bottom Reinforcement Along L:

Critical Analysis Load Combination	=	8	
Critical Load Combination	=	[9] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)	
Pu	=	328.46	kN
Mux	=	84.22	kNm
Muy	=	5.33	kNm
P/A	=	138.01	kN/sqm
Mx/Zx	=	124.89	kN/sqm
My/Zy	=	9.6	kN/sqm
Deff	=	317	mm
Beff	=	1400	mm
SPu	=	262.9	kN/sqm
Mu	=	SPu X B X Loff X Loff / 2	
	=	62.98	kNm
Pt	=	0.142	%
Ast Rqd (BM)	=	630	sqmm
Ast Prv (Distributed Across Total Width)	=	6 - T12 @ 280	
	=	679	sqmm

Bottom Reinforcement Along B:

Critical Analysis Load Combination	=	7	
Critical Load Combination	=	[8] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)	
Pu	=	350.82	kN
Mux	=	13.26	kNm
Muy	=	52.33	kNm
P/A	=	147.4	kN/sqm
Mx/Zx	=	19.66	kN/sqm
My/Zy	=	94.23	kN/sqm
Deff	=	301	mm
Leff	=	1700	mm
SPu	=	241.63	kN/sqm
Mu	=	SPu X L X Boff X Boff / 2	
	=	70.29	kNm
Pt	=	0.15	%
Ast Rqd (BM)	=	765	sqmm
Ast Prv (Distributed Across Total Length)	=	7 - T12 @ 285	
	=	792	sqmm

Design For Shear:

One Way Shear Along L:

Critical Section @ d from Column Face			
	=	317	mm
Critical Analysis Load Combination	=	8	
Critical Load Combination	=	[9] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)	
Pu	=	328.46	kN
Mux	=	84.22	kNm
Muy	=	5.33	kNm
P/A	=	138.01	kN/sqm
Mx/Zx	=	124.89	kN/sqm
My/Zy	=	9.6	kN/sqm
Deff	=	317	mm
Beff	=	1400	mm
SPu	=	262.9	kN/sqm
Vu	=	SPu X (Loff - d) X B	
	=	98.64	kN
Tv	=	Vu / (Beff X Deff)	
	=	0.22	N/sqmm

Tc	=	0.28	N/sqmm
Tv	<	Tc	

One Way Shear Along B:

Critical Section @ d from Column Face

	=	301	mm
Critical Analysis Load Combination	=	7	
Critical Load Combination	=	[8] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)	
Pu	=	350.82	kN
Mux	=	13.26	kNm
Muy	=	52.33	kNm
P/A	=	147.4	kN/sqm
Mx/Zx	=	19.66	kN/sqm
My/Zy	=	94.23	kN/sqm
Deff	=	301	mm
Leff	=	1700	mm
SPu	=	241.63	kN/sqm
Vu	=	SPu X (Boff - d) X L	
	=	116.66	kN
Tv	=	Vu / (Leff X Deff)	
	=	0.23	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

Design For Punching Shear:

Critical Section @ d/2 from Column Face

	=	154	mm
Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	438.89	kN
Mux	=	16.39	kNm
Muy	=	7.7	kNm
P/A	=	184.41	kN/sqm
Mx/Zx	=	24.3	kN/sqm
My/Zy	=	13.87	kN/sqm
Deff	=	309	mm
Leff	=	839	mm
Beff	=	539	mm

Punching Perimeter	=	2756	mm
Punching Area	=	851604	sqmm
SPu	=	Average Pressure	
	=	184.41	kN/sqm
Vu	=	SPu X ((L X B)-(Leff X Beff))	
	=	355.49	kN
Tv	=	Vu / Punching Area	
	=	0.417	N/sqmm
Tc	=	1.167	N/sqmm
Tv	<	Tc	

Load Transfer Check For Load Transfer From Column To Footing

Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	438.89	kN
A2	=	0.12	sqm
A1	=	3.51	sqm
Base Area	=	2.38	sqm
A1	>	Base Area	
Thus, A1	=	2.38	
Modification Factor	=	SquareRoot(A1/A2) < = 2	
SquareRoot(A1/A2)	=	4.4535	
Thus, Modification Factor	=	2	
Concrete Bearing Capacity	=	0.45 X Fck X Modification Factor X Column Area	
	=	2742.75	kN
Concrete Bearing Capacity	>	Pu, Hence Safe.	

Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Footing No	:	FC9	
Column No	:	C9	(230 mm X 450 mm)
Analysis No	:	51	
Concrete Grade	:	M25	N/sqmm
Steel Grade	:	Fe550	N/sqmm
Clear Cover	:	50	mm
Df	:	1.5 M	
Dw	:	4 M	
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%

Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Permissible area of loss of contact	=	0	%

Design cross section by : Average pressure

Footing Type : Pad

Footing Size (L x B x D) : 1800mm X 1600mm X 425mm

Without considering Effect of Water Table

Footing Self Wt., F1	=	30.60	kN
Soil Weight, S1	=	0.00	kN
Total Weight, W1	=	F1 + S1	
	=	30.6	kN
Offset Along L (Loff)	=	675	mm
Offset Along B (Boff)	=	685	mm

Check For Maximum Soil Pressure:

Critical Load Combination	=	[2] : (LOAD 3: DL) +(LOAD 4: LL) +(LOAD 1: EQ X)	
Pcomb	=	527.58	kN
P	=	Pcomb + W1	
P	=	558.18	kN
Mx	=	6.7	kNm
My	=	26.69	kNm
P/A	=	193.81	kN/sqm
Mx/Zx	=	7.75	kN/sqm
My/Zy	=	34.76	kN/sqm
Maximum Soil Pressure	=	236.32	kN/sqm
Allowable Soil Pressure	=	1.25x200	kN/sqm
	=	250	kN/sqm

Check For Minimum Soil Pressure:

Critical Load Combination	=	[3] : (LOAD 3: DL) + (LOAD 4: LL) -(LOAD 1: EQ X)	
Pcomb	=	366.99	kN
P	=	Pcomb + W1	
P	=	397.59	kN
Mx	=	12.38	kNm
My	=	-29.13	kNm
P/A	=	138.05	kN/sqm
Mx/Zx	=	14.33	kN/sqm

My/Zy	=	-37.93	kN/sqm
Minimum Soil Pressure	=	85.79	kN/sqm
	>	0	

Design For Bending:

Bottom Reinforcement Along L:

Critical Analysis Load Combination	=	8	
Critical Load Combination	=	[9] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)	
Pu	=	508.16	kN
Mux	=	70.1	kNm
Muy	=	-1.71	kNm
P/A	=	176.44	kN/sqm
Mx/Zx	=	81.14	kN/sqm
My/Zy	=	-2.22	kN/sqm
Deff	=	367	mm
Beff	=	1600	mm
SPu	=	257.58	kN/sqm
Mu	=	SPu X B X Loff X Loff / 2	
	=	93.89	kNm
Pt	=	0.139	%
Ast Rqd (BM)	=	816	sqmm
Ast Prv (Distributed Across Total Width)	=	8 - T12 @ 230	
	=	905	sqmm

Bottom Reinforcement Along B:

Critical Analysis Load Combination	=	7	
Critical Load Combination	=	[8] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)	
Pu	=	633.1	kN
Mux	=	8.04	kNm
Muy	=	32.03	kNm
P/A	=	219.83	kN/sqm
Mx/Zx	=	9.3	kN/sqm
My/Zy	=	41.71	kN/sqm
Deff	=	351	mm
Leff	=	1800	mm
SPu	=	261.53	kN/sqm
Mu	=	SPu X L X Boff X Boff / 2	
	=	110.45	kNm
Pt	=	0.145	%

Ast Rqd (BM)	=	918	sqmm
Ast Prv (Distributed Across Total Length)	=	9 - T12 @ 225	
	=	1018	sqmm

Design For Shear:

One Way Shear Along L:

Critical Section @ d from Column Face

	=	367	mm
Critical Analysis Load Combination	=	8	
Critical Load Combination	=	[9] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)	
Pu	=	508.16	kN
Mux	=	70.1	kNm
Muy	=	-1.71	kNm
P/A	=	176.44	kN/sqm
Mx/Zx	=	81.14	kN/sqm
My/Zy	=	-2.22	kN/sqm
Deff	=	367	mm
Beff	=	1600	mm
SPu	=	257.58	kN/sqm
Vu	=	SPu X (Loff - d) X B	
	=	126.94	kN
Tv	=	Vu / (Beff X Deff)	
	=	0.22	N/sqmm
Tc	=	0.28	N/sqmm
Tv	<	Tc	

One Way Shear Along B:

Critical Section @ d from Column Face

	=	351	mm
Critical Analysis Load Combination	=	7	
Critical Load Combination	=	[8] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)	
Pu	=	633.1	kN
Mux	=	8.04	kNm
Muy	=	32.03	kNm
P/A	=	219.83	kN/sqm
Mx/Zx	=	9.3	kN/sqm
My/Zy	=	41.71	kN/sqm

Deff	=	351	mm
Leff	=	1800	mm
SPu	=	261.53	kN/sqm
Vu	=	SPu X (Boff - d) X L	
	=	157.23	kN
Tv	=	Vu / (Leff X Deff)	
	=	0.25	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

Design For Punching Shear:

Critical Section @ d/2 from Column Face

	=	180	mm

Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	670.93	kN
Mux	=	14.31	kNm
Muy	=	-1.83	kNm
P/A	=	232.96	kN/sqm
Mx/Zx	=	16.56	kN/sqm
My/Zy	=	-2.38	kN/sqm

Deff	=	359	mm
Leff	=	809	mm
Beff	=	589	mm
Punching Perimeter	=	2796	mm
Punching Area	=	1003764	sqmm
SPu	=	Average Pressure	
	=	232.96	kN/sqm
Vu	=	SPu X ((L X B)-(Leff X Beff))	
	=	559.92	kN
Tv	=	Vu / Punching Area	
	=	0.558	N/sqmm
Tc	=	1.25	N/sqmm
Tv	<	Tc	

Load Transfer Check For Load Transfer From Column To Footing

Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	670.93	kN

A2	=	0.1	sqm
A1	=	4.15	sqm
Base Area	=	2.88	sqm
A1	>	Base Area	
Thus, A1	=	2.88	
Modification Factor	=	SquareRoot(A1/A2) < = 2	
SquareRoot(A1/A2)	=	5.3666	
Thus, Modification Factor	=	2	
Concrete Bearing Capacity	=	0.45 X Fck X Modification Factor X Column Area	
	=	2328.75	kN
Concrete Bearing Capacity	>	Pu, Hence Safe.	

Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Footing No	:	FC10	
Column No	:	C10	(230 mm X 450 mm)
Analysis No	:	69	
Concrete Grade	:	M25	N/sqmm
Steel Grade	:	Fe550	N/sqmm
Clear Cover	:	50	mm
Df	:	1.5 M	
Dw	:	4 M	
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Permissible area of loss of contact	=	0	%
Design cross section by	:	Average pressure	
Footing Type	:	Pad	
Footing Size (L x B x D)	:	1700mm X 1500mm X 425mm	

Without considering Effect of Water Table

Footing Self Wt., F1	=	27.09	kN
Soil Weight, S1	=	0.00	kN
Total Weight, W1	=	F1 + S1	
	=	27.09	kN
Offset Along L (Loff)	=	625	mm

Offset Along B (Boff)	=	635	mm

Check For Maximum Soil Pressure:

Critical Load Combination	=	[3] : (LOAD 3: DL) +(LOAD 4: LL) -(LOAD 1: EQ X)	
Pcomb	=	455.41	kN
P	=	Pcomb + W1	
P	=	482.50	kN
Mx	=	-0.88	kNm
My	=	-30.57	kNm
P/A	=	189.22	kN/sqm
Mx/Zx	=	-1.22	kN/sqm
My/Zy	=	-47.95	kN/sqm
Maximum Soil Pressure	=	238.39	kN/sqm
Allowable Soil Pressure	=	1.25x200	kN/sqm
	=	250	kN/sqm

Check For Minimum Soil Pressure:

Critical Load Combination	=	[2] : (LOAD 3: DL) + (LOAD 4: LL) +(LOAD 1: EQ X)	
Pcomb	=	278.16	kN
P	=	Pcomb + W1	
P	=	305.26	kN
Mx	=	6	kNm
My	=	29.35	kNm
P/A	=	119.71	kN/sqm
Mx/Zx	=	8.3	kN/sqm
My/Zy	=	46.03	kN/sqm
Minimum Soil Pressure	=	65.37	kN/sqm
	>	0	

Design For Bending:

Bottom Reinforcement Along L:

Critical Analysis Load Combination	=	13	
Critical Load Combination	=	[14] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	
Pu	=	487.18	kN
Mux	=	48.65	kNm
Muy	=	-3.26	kNm
P/A	=	191.05	kN/sqm
Mx/Zx	=	67.34	kN/sqm
My/Zy	=	-5.12	kN/sqm

Deff	=	367	mm
Beff	=	1500	mm
SPu	=	258.39	kN/sqm
Mu	=	SPu X B X Loff X Loff / 2	
	=	75.7	kNm
Pt	=	0.139	%
Ast Rqd (BM)	=	765	sqmm
Ast Prv (Distributed Across Total Width)	=	7 - T12 @ 250	
	=	792	sqmm

Bottom Reinforcement Along B:

Critical Analysis Load Combination	=	14	
Critical Load Combination	=	[15] : 1.5 (LOAD 3: DL) -1.5 (LOAD 1: EQ X)	
Pu	=	534.63	kN
Mux	=	-2.61	kNm
Muy	=	-45.15	kNm
P/A	=	209.66	kN/sqm
Mx/Zx	=	-3.61	kN/sqm
My/Zy	=	-70.82	kN/sqm

Deff	=	351	mm
Leff	=	1700	mm
SPu	=	280.48	kN/sqm
Mu	=	SPu X L X Boff X Boff / 2	
	=	96.13	kNm
Pt	=	0.145	%
Ast Rqd (BM)	=	867	sqmm
Ast Prv (Distributed Across Total Length)	=	8 - T12 @ 245	
	=	905	sqmm

Design For Shear:

One Way Shear Along L:

Critical Section @ d from Column Face			
	=	367	mm
Critical Analysis Load Combination	=	13	
Critical Load Combination	=	[14] : 1.5 (LOAD 3: DL) +1.5 (LOAD 2: EQ Z)	
Pu	=	487.18	kN
Mux	=	48.65	kNm
Muy	=	-3.26	kNm
P/A	=	191.05	kN/sqm
Mx/Zx	=	67.34	kN/sqm

My/Zy	=	-5.12	kN/sqm
Deff	=	367	mm
Beff	=	1500	mm
SPu	=	258.39	kN/sqm
Vu	=	SPu X (Loff - d) X B	
	=	100.00	kN
Tv	=	Vu / (Beff X Deff)	
	=	0.18	N/sqmm
Tc	=	0.28	N/sqmm
Tv	<	Tc	

One Way Shear Along B:

Critical Section @ d from Column Face

	=	351	mm
Critical Analysis Load Combination	=	14	
Critical Load Combination	=	[15] : 1.5 (LOAD 3: DL) - 1.5 (LOAD 1: EQ X)	
Pu	=	534.63	kN
Mux	=	-2.61	kNm
Muy	=	-45.15	kNm
P/A	=	209.66	kN/sqm
Mx/Zx	=	-3.61	kN/sqm
My/Zy	=	-70.82	kN/sqm
Deff	=	351	mm
Leff	=	1700	mm
SPu	=	280.48	kN/sqm
Vu	=	SPu X (Boff - d) X L	
	=	135.42	kN
Tv	=	Vu / (Leff X Deff)	
	=	0.23	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

Design For Punching Shear:

Critical Section @ d/2 from Column Face

	=	180	mm
Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	550.18	kN

Mux	=	3.84	kNm
Muy	=	-0.92	kNm
P/A	=	215.76	kN/sqm
Mx/Zx	=	5.31	kN/sqm
My/Zy	=	-1.44	kN/sqm
Deff	=	359	mm
Leff	=	809	mm
Beff	=	589	mm
Punching Perimeter	=	2796	mm
Punching Area	=	1003764	sqmm
SPu	=	Average Pressure	
	=	215.76	kN/sqm
Vu	=	SPu X ((L X B)-(Leff X Beff))	
	=	447.37	kN
Tv	=	Vu / Punching Area	
	=	0.446	N/sqmm
Tc	=	1.25	N/sqmm
Tv	<	Tc	

Load Transfer Check For Load Transfer From Column To Footing

Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	550.18	kN
A2	=	0.1	sqm
A1	=	4.15	sqm
Base Area	=	2.55	sqm
A1	>	Base Area	
Thus, A1	=	2.55	
Modification Factor	=	SquareRoot(A1/A2) < = 2	
SquareRoot(A1/A2)	=	5.0498	
Thus, Modification Factor	=	2	
Concrete Bearing Capacity	=	0.45 X Fck X Modification Factor X Column Area	
	=	2328.75	kN
Concrete Bearing Capacity	>	Pu, Hence Safe.	

Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Footing No	:	FC11	
Column No	:	C11	(230 mm X 530 mm)
Analysis No	:	70	

Concrete Grade	:	M25	N/sqmm
Steel Grade	:	Fe550	N/sqmm
Clear Cover	:	50	mm
Df	:	1.5 M	
Dw	:	4 M	

Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Permissible area of loss of contact	=	0	%

Design cross section by : Average pressure

Footing Type : Pad

Footing Size (L x B x D) : 1950mm X 1650mm X 425mm

Without considering Effect of Water Table

Footing Self Wt., F1	=	34.19	kN
Soil Weight, S1	=	0.00	kN
Total Weight, W1	=	F1 + S1	
	=	34.19	kN
Offset Along L (Loff)	=	710	mm
Offset Along B (Boff)	=	710	mm

Check For Maximum Soil Pressure:

Critical Load Combination	=	[4] : (LOAD 3: DL) +(LOAD 4: LL) +(LOAD 2: EQ Z)	
Pcomb	=	561.21	kN
P	=	Pcomb + W1	
P	=	595.40	kN
Mx	=	54.6	kNm
My	=	2.46	kNm
P/A	=	185.05	kN/sqm
Mx/Zx	=	52.22	kN/sqm
My/Zy	=	2.78	kN/sqm
Maximum Soil Pressure	=	240.05	kN/sqm
Allowable Soil Pressure	=	1.25x200	kN/sqm
	=	250	kN/sqm

Check For Minimum Soil Pressure:

Critical Load Combination	=	[5] : (LOAD 3: DL) + (LOAD 4: LL) -(LOAD 2: EQ Z)	
Pcomb	=	439.60	kN
P	=	Pcomb + W1	
P	=	473.79	kN
Mx	=	-53.01	kNm
My	=	7.61	kNm
P/A	=	147.25	kN/sqm
Mx/Zx	=	-50.7	kN/sqm
My/Zy	=	8.6	kN/sqm
Minimum Soil Pressure	=	87.96	kN/sqm
	>	0	

Design For Bending:

Bottom Reinforcement Along L:

Critical Analysis Load Combination	=	8	
Critical Load Combination	=	[9] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)	
Pu	=	673.45	kN
Mux	=	65.52	kNm
Muy	=	2.95	kNm
P/A	=	209.31	kN/sqm
Mx/Zx	=	62.66	kN/sqm
My/Zy	=	3.34	kN/sqm
Deff	=	367	mm
Beff	=	1650	mm
SPu	=	271.97	kN/sqm
Mu	=	SPu X B X Loff X Loff / 2	
	=	113.11	kNm
Pt	=	0.139	%
Ast Rqd (BM)	=	842	sqmm
Ast Prv (Distributed Across Total Width)	=	8 - T12 @ 240	
	=	905	sqmm

Bottom Reinforcement Along B:

Critical Analysis Load Combination	=	7	
Critical Load Combination	=	[8] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)	
Pu	=	604.09	kN
Mux	=	1.43	kNm
Muy	=	54.63	kNm
P/A	=	187.75	kN/sqm

Mx/Zx	=	1.37	kN/sqm
My/Zy	=	61.74	kN/sqm
Deff	=	351	mm
Leff	=	1950	mm
SPu	=	249.5	kN/sqm
Mu	=	SPu X L X Boff X Boff / 2	
	=	122.63	kNm
Pt	=	0.145	%
Ast Rqd (BM)	=	995	sqmm
Ast Prv (Distributed Across Total Length)	=	9 - T12 @ 245	
	=	1018	sqmm

Design For Shear:

One Way Shear Along L:

Critical Section @ d from Column Face

	=	367	mm
Critical Analysis Load Combination	=	8	
Critical Load Combination	=	[9] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)	
Pu	=	673.45	kN
Mux	=	65.52	kNm
Muy	=	2.95	kNm
P/A	=	209.31	kN/sqm
Mx/Zx	=	62.66	kN/sqm
My/Zy	=	3.34	kN/sqm
Deff	=	367	mm
Beff	=	1650	mm
SPu	=	271.97	kN/sqm
Vu	=	SPu X (Loff - d) X B	
	=	153.92	kN
Tv	=	Vu / (Beff X Deff)	
	=	0.25	N/sqmm
Tc	=	0.28	N/sqmm
Tv	<	Tc	

One Way Shear Along B:

Critical Section @ d from Column Face

	=	351	mm
Critical Analysis Load Combination	=	7	

Critical Load Combination	=	[8] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)	
Pu	=	604.09	kN
Mux	=	1.43	kNm
Muy	=	54.63	kNm
P/A	=	187.75	kN/sqm
Mx/Zx	=	1.37	kN/sqm
My/Zy	=	61.74	kN/sqm
Deff	=	351	mm
Leff	=	1950	mm
SPu	=	249.5	kN/sqm
Vu	=	SPu X (Boff - d) X L	
	=	174.66	kN
Tv	=	Vu / (Leff X Deff)	
	=	0.26	N/sqmm
Tc	=	0.29	N/sqmm
Tv	<	Tc	

Design For Punching Shear:

Critical Section @ d/2 from Column Face

	=	180	mm
Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	750.61	kN
Mux	=	1.19	kNm
Muy	=	7.55	kNm
P/A	=	233.29	kN/sqm
Mx/Zx	=	1.14	kN/sqm
My/Zy	=	8.53	kN/sqm
Deff	=	359	mm
Leff	=	889	mm
Beff	=	589	mm
Punching Perimeter	=	2956	mm
Punching Area	=	1061204	sqmm
SPu	=	Average Pressure	
	=	233.29	kN/sqm
Vu	=	SPu X ((L X B)-(Leff X Beff))	
	=	628.46	kN
Tv	=	Vu / Punching Area	

	=	0.592	N/sqmm
Tc	=	1.167	N/sqmm
Tv	<	Tc	

Load Transfer Check For Load Transfer From Column To Footing

Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	750.61	kN
A2	=	0.12	sqm
A1	=	4.3	sqm
Base Area	=	3.22	sqm
A1	>	Base Area	
Thus, A1	=	3.22	
Modification Factor	=	SquareRoot(A1/A2) < = 2	
SquareRoot(A1/A2)	=	5.1801	
Thus, Modification Factor	=	2	
Concrete Bearing Capacity	=	0.45 X Fck X Modification Factor X Column Area	
	=	2742.75	kN
Concrete Bearing Capacity	>	Pu, Hence Safe.	

Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Footing No	:	FC12	
Column No	:	C12	(230 mm X 450 mm)
Analysis No	:	71	
Concrete Grade	:	M25	N/sqmm
Steel Grade	:	Fe550	N/sqmm
Clear Cover	:	50	mm
Df	:	1.5 M	
Dw	:	4 M	
Density of Soil	=	18	kN/cum
Soil Bearing Capacity	=	200	kN/sqm
Permissible SBC Increase for EQ	=	25	%
Permissible SBC Increase for Wind	=	25	%
Live Load Reduction	=	0	%
Permissible area of loss of contact	=	0	%
Design cross section by	:	Average pressure	

Footing Type	:	Pad	
Footing Size (L x B x D)	:	1900mm X 1700mm X 450mm	
Without considering Effect of Water Table			
Footing Self Wt., F1	=	36.34	kN
Soil Weight, S1	=	0.00	kN
Total Weight, W1	=	F1 + S1	
	=	36.34	kN
Offset Along L (Loff)	=	725	mm
Offset Along B (Boff)	=	735	mm

Check For Maximum Soil Pressure:

Critical Load Combination	=	[4] : (LOAD 3: DL) +(LOAD 4: LL) +(LOAD 2: EQ Z)	
Pcomb	=	610.51	kN
P	=	Pcomb + W1	
P	=	646.84	kN
Mx	=	45.28	kNm
My	=	-2.62	kNm
P/A	=	200.26	kN/sqm
Mx/Zx	=	44.27	kN/sqm
My/Zy	=	-2.86	kN/sqm
Maximum Soil Pressure	=	247.39	kN/sqm
Allowable Soil Pressure	=	1.25x200	kN/sqm
	=	250	kN/sqm

Check For Minimum Soil Pressure:

Critical Load Combination	=	[5] : (LOAD 3: DL) + (LOAD 4: LL) -(LOAD 2: EQ Z)	
Pcomb	=	467.31	kN
P	=	Pcomb + W1	
P	=	503.65	kN
Mx	=	-42.58	kNm
My	=	2.84	kNm
P/A	=	155.93	kN/sqm
Mx/Zx	=	-41.63	kN/sqm
My/Zy	=	3.11	kN/sqm
Minimum Soil Pressure	=	111.19	kN/sqm
	>	0	

Design For Bending:

Bottom Reinforcement Along L:

Critical Analysis Load Combination	=	8	

Critical Load Combination	=	[9] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)	
Pu	=	732.61	kN
Mux	=	54.33	kNm
Muy	=	-3.14	kNm
P/A	=	226.81	kN/sqm
Mx/Zx	=	53.12	kN/sqm
My/Zy	=	-3.44	kN/sqm
Deff	=	392	mm
Beff	=	1700	mm
SPu	=	279.93	kN/sqm
Mu	=	SPu X B X Loff X Loff / 2	
	=	125.07	kNm
Pt	=	0.138	%
Ast Rqd (BM)	=	918	sqmm
Ast Prv (Distributed Across Total Width)	=	9 - T12 @ 215	
	=	1018	sqmm

Bottom Reinforcement Along B:

Critical Analysis Load Combination	=	7	
Critical Load Combination	=	[8] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)	
Pu	=	748.46	kN
Mux	=	-2.99	kNm
Muy	=	36.73	kNm
P/A	=	231.72	kN/sqm
Mx/Zx	=	-2.92	kN/sqm
My/Zy	=	40.14	kN/sqm
Deff	=	376	mm
Leff	=	1900	mm
SPu	=	271.86	kN/sqm
Mu	=	SPu X L X Boff X Boff / 2	
	=	139.52	kNm
Pt	=	0.144	%
Ast Rqd (BM)	=	1026	sqmm
Ast Prv (Distributed Across Total Length)	=	10 - T12 @ 215	
	=	1131	sqmm

Design For Shear:

One Way Shear Along L:

Critical Section @ d from Column Face			
	=	392	mm
Critical Analysis Load Combination	=	8	
Critical Load Combination	=	[9] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 2: EQ Z)	
Pu	=	732.61	kN
Mux	=	54.33	kNm
Muy	=	-3.14	kNm
P/A	=	226.81	kN/sqm
Mx/Zx	=	53.12	kN/sqm
My/Zy	=	-3.44	kN/sqm
Deff	=	392	mm
Beff	=	1700	mm
SPu	=	279.93	kN/sqm
Vu	=	SPu X (Loff - d) X B	
	=	158.47	kN
Tv	=	Vu / (Beff X Deff)	
	=	0.24	N/sqmm
Tc	=	0.28	N/sqmm
Tv	<	Tc	

One Way Shear Along B:

Critical Section @ d from Column Face			
	=	376	mm
Critical Analysis Load Combination	=	7	
Critical Load Combination	=	[8] : 1.2 (LOAD 3: DL) +1.2 (LOAD 4: LL) +1.2 (LOAD 1: EQ X)	
Pu	=	748.46	kN
Mux	=	-2.99	kNm
Muy	=	36.73	kNm
P/A	=	231.72	kN/sqm
Mx/Zx	=	-2.92	kN/sqm
My/Zy	=	40.14	kN/sqm
Deff	=	376	mm
Leff	=	1900	mm
SPu	=	271.86	kN/sqm
Vu	=	SPu X (Boff - d) X L	
	=	185.44	kN
Tv	=	Vu / (Leff X Deff)	
	=	0.26	N/sqmm

Tc	=	0.29	N/sqmm
Tv	<	Tc	

Design For Punching Shear:

Critical Section @ d/2 from Column Face

	=	192	mm

Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	808.36	kN
Mux	=	2.02	kNm
Muy	=	0.17	kNm
P/A	=	250.27	kN/sqm
Mx/Zx	=	1.98	kN/sqm
My/Zy	=	0.18	kN/sqm
Deff	=	384	mm
Leff	=	834	mm
Beff	=	614	mm
Punching Perimeter	=	2896	mm
Punching Area	=	1112064	sqmm
SPu	=	Average Pressure	
	=	250.27	kN/sqm
Vu	=	SPu X ((L X B)-(Leff X Beff))	
	=	680.21	kN
Tv	=	Vu / Punching Area	
	=	0.612	N/sqmm
Tc	=	1.25	N/sqmm
Tv	<	Tc	

Load Transfer Check For Load Transfer From Column To Footing

Critical Analysis Load Combination	=	5	
Critical Load Combination	=	[6] : 1.5 (LOAD 3: DL) +1.5 (LOAD 4: LL)	
Pu	=	808.36	kN
A2	=	0.1	sqm
A1	=	4.57	sqm
Base Area	=	3.23	sqm
A1	>	Base Area	
Thus, A1	=	3.23	
Modification Factor	=	SquareRoot(A1/A2) < = 2	
SquareRoot(A1/A2)	=	5.6833	
Thus, Modification Factor	=	2	

Concrete Bearing Capacity	=	0.45 X Fck X Modification Factor X Column Area
	=	2328.75 kN
Concrete Bearing Capacity	>	Pu, Hence Safe.